



香港大學

THE UNIVERSITY OF HONG KONG

CIVIL 7023

Industrialized Construction with Automation and Robotics

03 September 2026

Prof. LI Xiao

Assistant Professor

Dept. of Civil Engineering

The University of Hong Kong

Pokfulam, Hong Kong

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Course lecturer and TAs for CIVL 7023

✿ Prof. LI Xiao

- ✿ Research Interests: Industrialized Construction, Construction Informatics, Construction Robotics

✿ Contacts:

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CI³ Lab

<https://cicubelab.civil.hku.hk/>



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Mr. QIAN Chen

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Teaching Assistant

Introduction to CIVIL 7023

- ❁ **Subject Code:** CIVIL 7023
- ❁ **Subject Title:** Industrialized Construction with Automation and Robotics
- ❁ **Class Timetable:** Thursday 7:00-9:20 pm (Week 1-4, 6, 8-13)
- ❁ **Venue:** KKLG 102, LG1/F, K.K. Leung Building;
- ❁ **Department:** Civil Engineering
- ❁ **Subject Lecturer:** X Li



Study schedule

Week	Course Content/Guest Speaker	Assessment	Contact Hours	Total Study Load (Hours)	Course Learning Outcomes
<i>Weeks 1-4, 6, 8-13 Lectures and tutorials by Prof. LI Xiao, Guest Speakers & TAs</i>					
1-03/09/2026	Introduction to Industrialized Construction	Assessment Brief	2.5	10	1,2,3,4,5
2-10/09/2026	Automatic Design for Industrialized Building		2.5	10	1,2,3,4,5
3-17/09/2026	<i>Automatic Quality Inspection in Production-Guest Lecture</i>	Project 1 Tutorial	2.5	10	1,2,3,4,5
4-24/09/2026	<i>Adaptive Human-Robot Production Scheduling-Guest Lecture</i>	Project 2 Tutorial	2.5	10	1,2,3,4,5
6-08/10/2026	<i>Assembly Robots-Guest Lecture</i>	Project 3 Tutorial	2.5	10	1,2,3,4,5
8-22/10/2026	Automatic & Robotic Manufacturing in Industrialized Construction		2.5	10	1,2,3,4,5
9-29/10/2026	Intelligent Supply Chain Management in Industrialized Construction		2.5	10	1,2,3,4,5
10-05/11/2026	On-site Robotic Assembly in Industrialized Construction		2.5	10	1,2,3,4,5
11-12/11/2026	Smart Work Packaging for Project Management		2.5	10	1,2,3,4,5
12-19/11/2026	Intelligent Maintenance for Industrialized Building		2.5	10	1,2,3,4,5
<i>Project Oral Presentation (Submission Deadline for Project Report: 30/11/2026)</i>					
13-26/11/2026	Project Presentation	Project Assessment	2.5	10	1,2,3,4,5
07/12-23/12/2025	4 descriptive questions (open-book exam)	Written Examination			

We will award top 3 winners for the project presentation. The presentation will be assessed by all lecturers in this course

Assessment plan

Assessment Name	Continuous Assessment + Written Examination
Description	You will understand the basic concepts, method, principles, and technologies behind the industrialized construction
Learning outcomes	The learning outcomes of the assessment relates to: 1. Understand the fundamentals of industrialized construction and its benefits for modern civil engineering projects; 2. Familiarize with various industrialized construction techniques and technologies 3. Develop problem-solving skills in applying industrialized construction methods to civil engineering projects 4. Analyze and optimize construction workflows using industrialized construction principles 5. Understand the challenges and future trends in industrialized construction
Format	Group project (Max. 4 students for a group)+ 4 descriptive questions (open-book exam)
Weight	<u>50% (Continuous Assessment, Project Report (30%)+Presentation(20%)) + 50% (2-hour written examination, 4 questions)</u>
Submission/examination date	Project report with your presentation slides submission: 30 Nov 2026; Project Presentation: 26 Nov 2026, 10 mins presentation & 5 mins Q&A;
Feedback/Questions	Through email or Moodle.
Criteria	(1) Demonstrate understanding and knowledge of industrialized construction principles and technical literature; (2) Demonstrate understanding and knowledge of problem analysis, analytical tools and techniques, and integrated/systems approach on industrialized construction; (3) Demonstrate understanding and knowledge of practical and workshop skills, teamwork, communication, sustainability, and equality for the projects in industrialized construction;

Please fill your group information (Names & Student No.) with project no. before **17 Sept 2026** via https://connecthkuhk-my.sharepoint.com/:x/g/personal/u3013967_connect_hku_hk/IQAikhEGY0soSJkmbU6YuJJqAZBSYGS46r1TkEzs-ifs-To?e=Jmw5up
Each project's quota will be limited to **6** groups (First come, first served). Presentation sequence will start from the group 1 in the project 1

Assessment I: Project report

- ✿ **Title of Project** (should be informative, but short and concise, 20 words maximum)
- ✿ **Abstract** (150-250 words)
- ✿ **Keywords** (a maximum of five keywords to characterize the work of the paper)
- ✿ **Main Body** (Maximum: 10,000 words)
 - ★ 1. Introduction
 - ★ 2. Literature Review
 - ★ 3. Research Methodology
 - ★ 4. Results and Discussion
 - ★ 5. Conclusions
- ✿ **Acknowledgements** (briefly acknowledge those who helped with the research)
- ✿ **References** (list all cited references but avoid a long list of references. The APA referencing style should be used)

Note: Font: *Times New Roman*, Font Size: 12 point, Margin: 2.5 cm all round, Spacing: Single spacing, Deadline for submission: **30 Nov 2026**

Submissions through Moodle (Turnitin similarity $\leq 10\%$, AI similarity $\leq 10\%$), excluding the references

Project-1

✿ Geometric Measurement of Module and Prefabricated Components for MiC Quality Inspection

(Dr. HONG Jie, jiehong@hku.hk; Dr. LI Tingtian, tingtian@hku.hk; Dr. CHEN Zhe, cz77@connect.hku.hk)

✿ Aim: To develop a geometric measurement framework for comprehensive MiC quality inspection purposes

✿ Objectives:

- ★ Literature review on deep 2D detection/segmentation, deep 3D detection/segmentation, and geometric measurement.
- ★ Identify the main drawbacks of existing geometric measurement methods as the motivation for the following algorithm design.
- ★ Design a task-aware dual-head deep-learning framework which can do the measurements for as-built MiC module and prefabricated components. Foundation-based models are welcome to use.
- ★ Implement the designed measurement framework, and validate it on the provided dataset. Compare the results with the state-of-the-art methods.

✿ Deliverables: a project report

✿ Requirements: introduction, related works, methodology (framework design), and experimental validation (result analysis and visualizations)

✿ Resources:

- ★ (i) Related datasets for MiC module and will update at <https://github.com/CI3LAB>.

Project-1

❁ Concrete Modular Integrated Construction (MiC) Automatic Quality Inspection



❑ Livingroom/bedroom



❑ Kitchen



❑ Bathroom



❑ Livingroom/bedroom+ bathroom



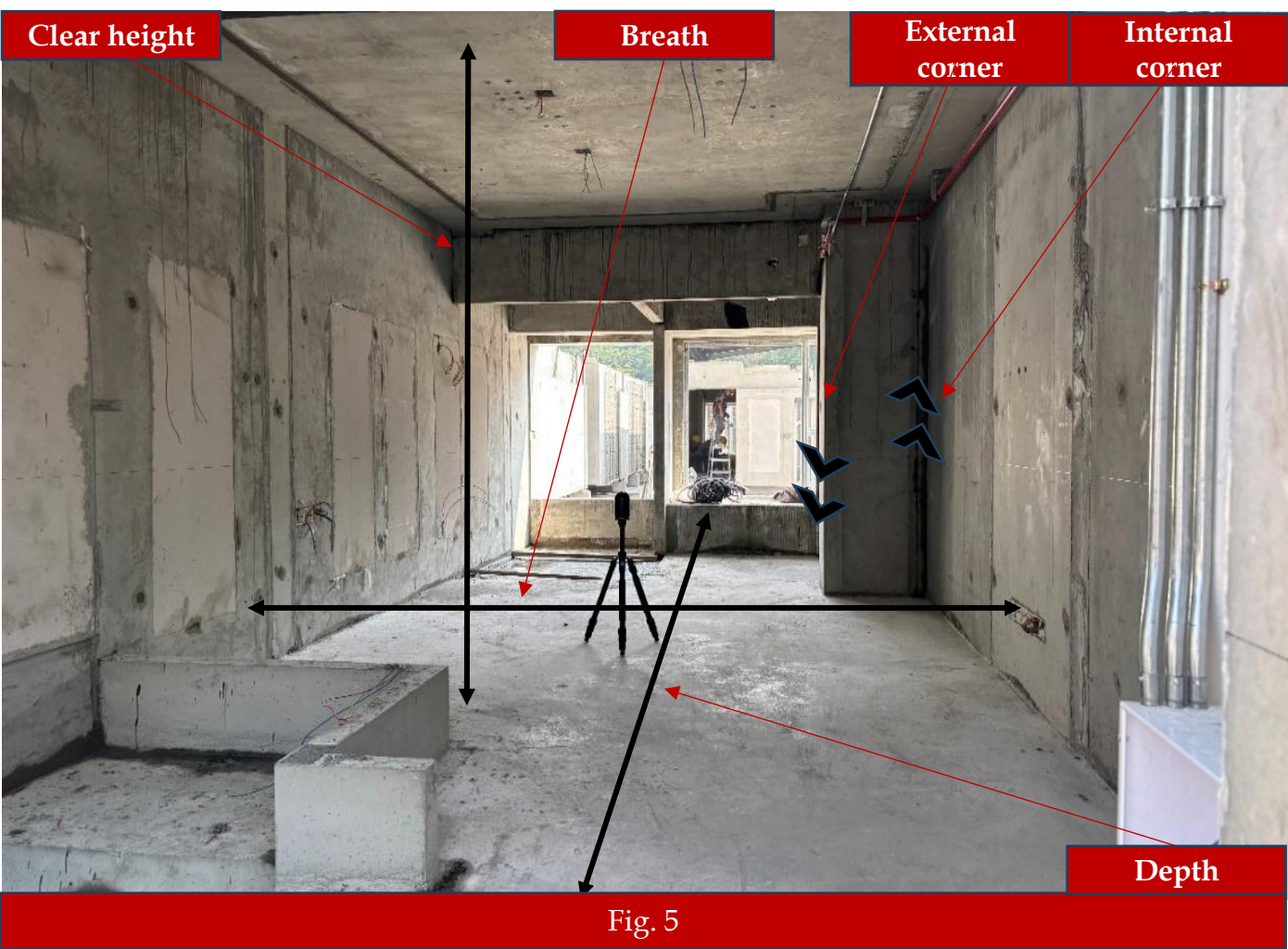
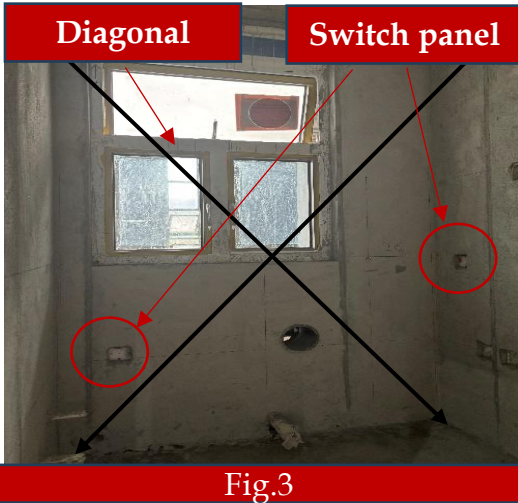
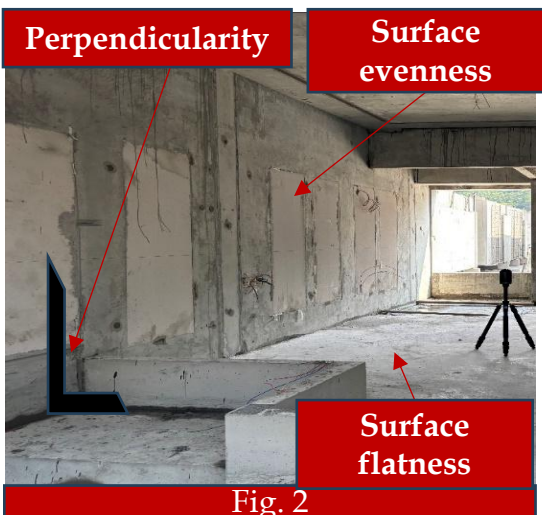
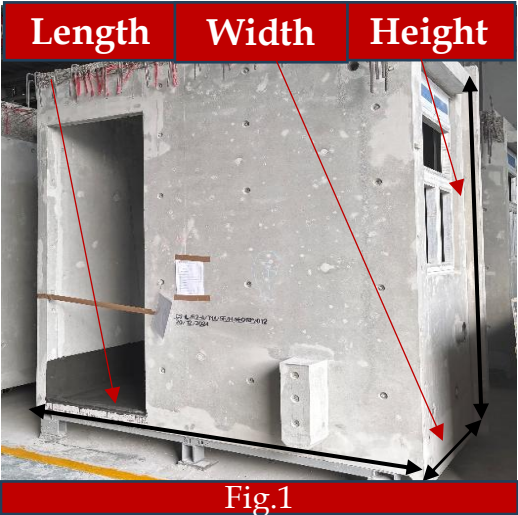
❑ Livingroom/bedroom+ kitchen



❑ Bathroom+ kitchen

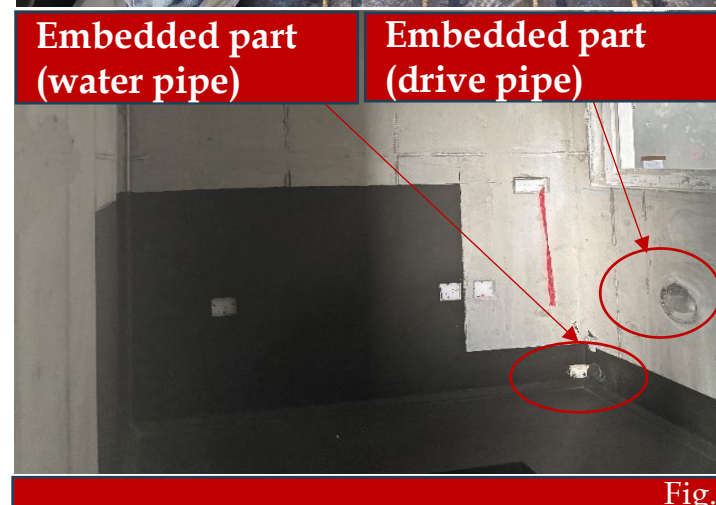
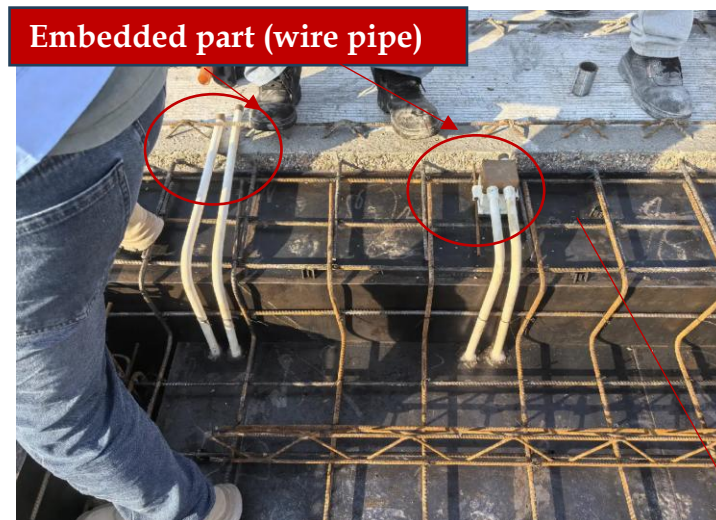
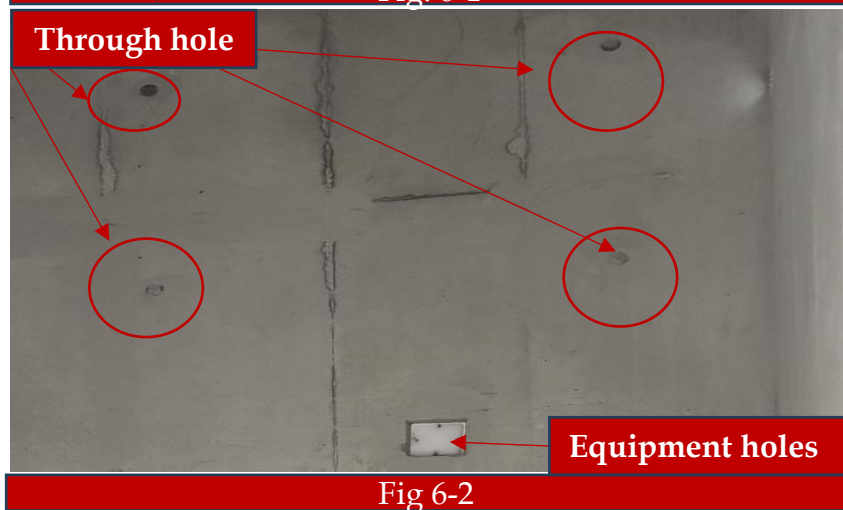
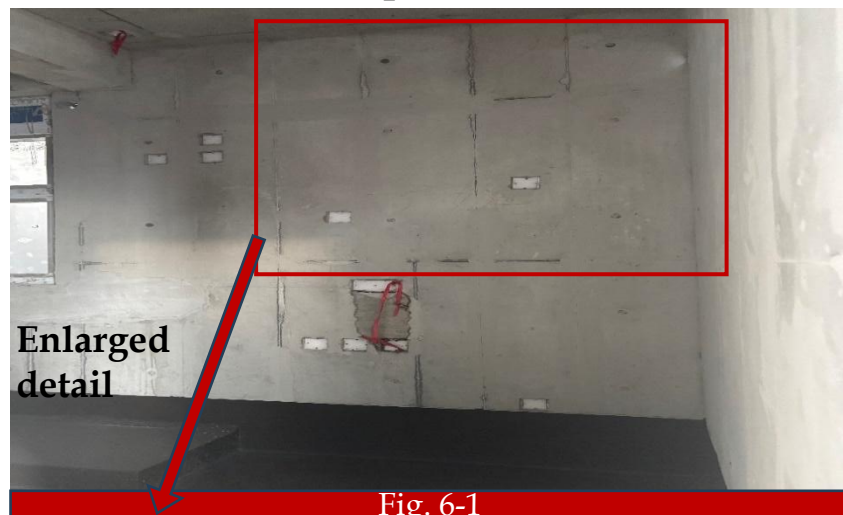
Project-1

Concrete MiC Inspection Items



Project-1

❁ Concrete MiC Inspection Items



Project-2

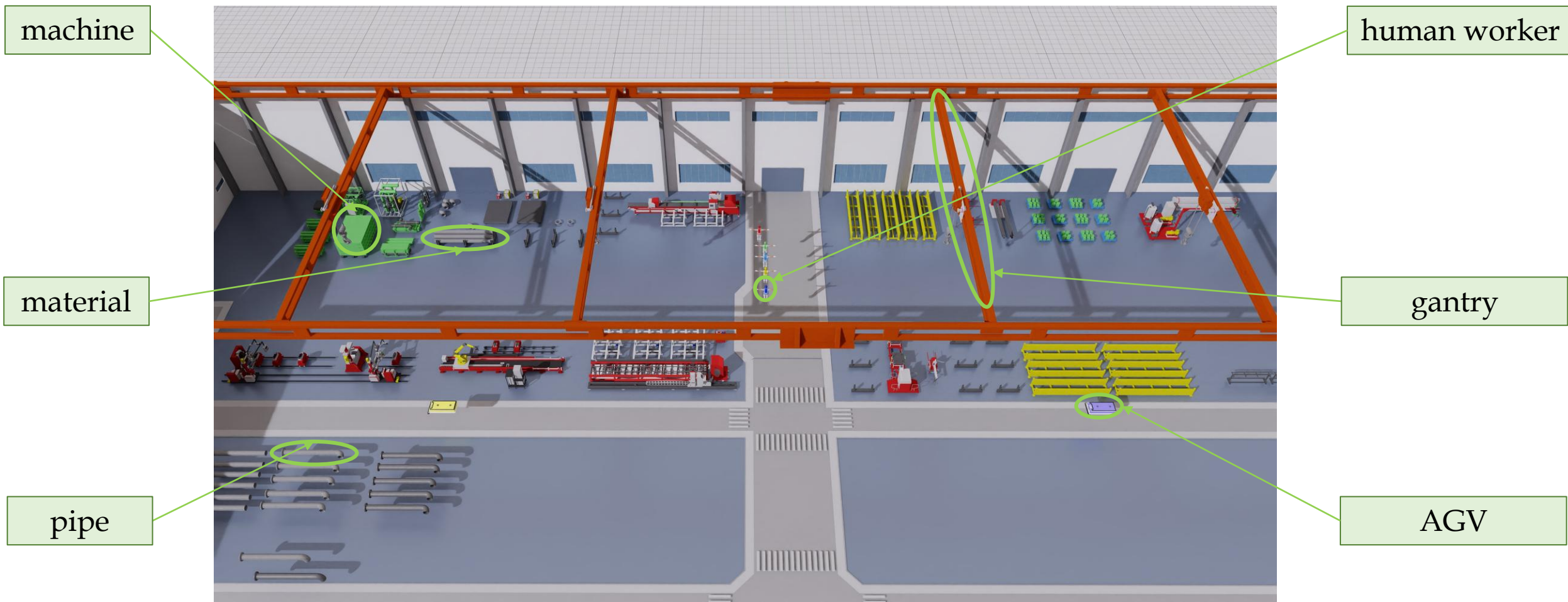
✿ Adaptive Human-Robot Production Scheduling for Resilient Modular Construction Manufacturing

(Dr. ZHENG Hao, hzheng26@hku.hk; Dr. ZHANG Yaning, zynhku@hku.hk)

- ✿ Aim: To develop and validate an adaptive human-robot scheduling framework for efficient and robust modular construction production under changing human, equipment, and order states
- ✿ Objectives:
 - ★ Formulate a dynamic multi-resource scheduling model for modular construction production, incorporating task precedence, resource constraints, human-robot collaboration modes, human fatigue dynamics, equipment failures, and urgent order arrivals or cancellations.
 - ★ Develop an Isaac Sim-based human-robot production environment representing human workers, robots, machines, gantry cranes, and production workflows, with controlled disruption injection and closed-loop schedule execution.
 - ★ Develop and evaluate an adaptive scheduling algorithm for online task scheduling and rescheduling using state feedback from the built Isaac Sim environment.
- ✿ Deliverables: a project report
- ✿ Requirements: Introduction, related works, methodology, experiments, and conclusion
- ✿ Resources:
 - ★ USD assets for a pipe production line, including humans, robots, machines, and gantry cranes.
 - ★ Isaac Sim Tutorial: <https://docs.isaacsim.omniverse.nvidia.com/6.0.0/index.html>.
 - ★ Google OR-Tool: <https://developers.google.com/optimization/scheduling>
 - ★ Pinedo, M. L. (2022). Scheduling: Theory, Algorithms, and Systems. Germany: Prentice-Hall.

Project-2

An example Isaac Sim environment for a pipe production factory featuring multiple workers, AGVs, machines, and gantry cranes.



Project-3

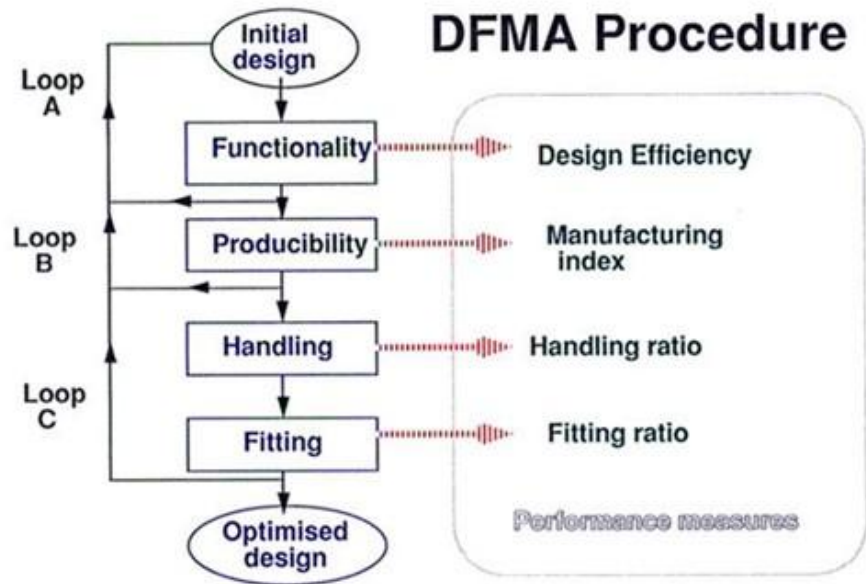
✿ Design and Robotic Assembly of Sintered Lunar-Regolith Components for In-Situ Lunar Construction

(Dr. BAI Songnan, snbai@hku.hk; Dr. SHI Xiongtao, xotaoshi@hku.hk; Dr. WANG Kun, kun7wang@hku.hk)

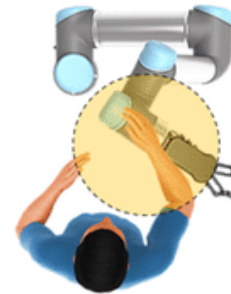
- ✿ Aim: Design a small lunar-base structural system and develop its robotic assembly process using a provided 6-DoF manipulator with gripper.
- ✿ Objectives:
 - ✿ Design a lunar-base structural unit, including component geometry, interfaces, and assembly sequence.
 - ✿ Determine feasible grasping and assembly poses using kinematics and Jacobian analysis.
 - ✿ Plan collision-free motions and refine the component design for improved reachability and alignment.
 - ✿ Implement a suitable controller, such as kinematic, hybrid position-force, or impedance/admittance control.
- ✿ Deliverables: A project report, component CAD models, simulation model, and demonstration video
- ✿ Requirements: Complete the assembly task in simulation and evaluate pose accuracy, trajectory feasibility, and basic joint torques under lunar gravity
- ✿ Resources:
 - ✿ A URDF model of a 6-DoF robotic manipulator with gripper

Assessment II: written examination (sample question)

- ✿ Explain the approach of Design for Manufacture and Assembly (DfMA) in Industrialized Construction. How does AI-driven DfMA influence the construction process and final product?
- ✿ Explain the principles of Lean manufacturing in Industrialized Construction. How to achieve human-robot collaboration-based Lean production for customized and resilient manufacturing processes?



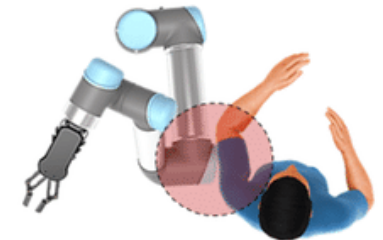
Safety monitored stop



Hand guiding



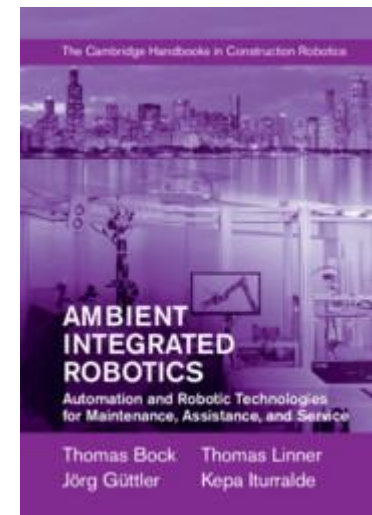
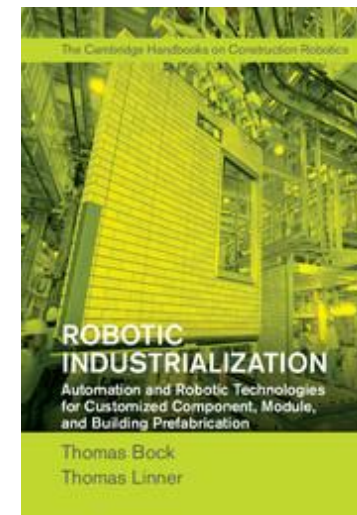
Speed and separation monitoring



Power and force limiting

Recommended readings/textbooks

- ✿ Li, X. (2026). *Industrialized Construction in the Age of AI and Robotics*. Taylor & Francis.
- ✿ Bock, T., & Linner, T. (2015). *Robotic industrialization: Automation and Robotic Technologies for Customized Component, Module, and Building Prefabrication*. Cambridge University Press.
- ✿ Bock, T., & Linner, T. (2017). *Construction Robots: Elementary Technologies and Single-Task Construction Robots*. Cambridge University Press.
- ✿ Bock, T., & Linner, T. (2019). *Ambient Integrated Robotics: Automation and Robotic Technologies for Maintenance, Assistance, and Service*. Cambridge University Press.





Thanks for your attention!

Q&A

Course Feedback

